9.6 Solving Right Triangles

Name _____ Period ____ Date _____

Geometry 9.6 · Student sheet

I can…

- use an inverse trigonometric ratio to find an angle measure.
- solve a right triangle by finding all of its side lengths and angle measures.
- solve real-life problems with right triangles.

Vocabulary · fill in the missing word

Inverse tangent — if tan A = _____, then $\tan^{-1} x = m\angle A$

Inverse sine and inverse cosine — $\sin^{-1}$ and $\cos^{-1}$: they give the angle _____ from a sine or cosine ratio

Solve a right triangle — find all of the unknown side lengths and angle _____

Part 1 · Discovery: the calculator works backward

Draw each right triangle. At the end of the longer leg, measure the acute angle with a protractor. Then find $\tan^{-1}$(opposite ÷ adjacent) on the calculator.

Legs	Opposite ÷ adjacent	Protractor	$\tan^{-1}$ on the calculator
5 cm and 7 cm			
3 cm and 8 cm			

$\tan^{-1}$ takes a _____ and gives an _____.

Part 2 · Notes and examples (fill in each note, then work the example)

Inverse trig ratios: if tan A = x, then _____ = m∠A. It works the same way with $\sin^{-1}$ and _____.

Example 1

Find m∠A to the nearest tenth of a degree.

a. tan A = 0.75

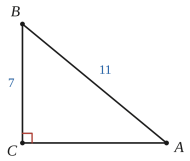
b. sin A = 0.5

c. cos A = 0.8

Which ratio? Use the two sides you _____: opposite and adjacent → tan, opposite and hypotenuse → _____, adjacent and hypotenuse → _____.

Example 2

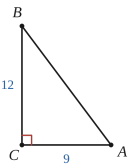
Find m∠A to the nearest tenth of a degree.



To solve a right triangle: find the third side with the _____ Theorem, one acute angle with an inverse ratio, and the other acute angle with 90° minus the first.

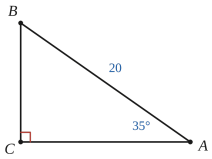
Example 3

Solve right △ABC: find AB, m∠A, and m∠B. Round to the nearest tenth.



Example 4

Solve right △ABC: find BC, AC, and m∠B. Round to the nearest tenth.



Example 5

A wheelchair ramp rises 2 feet over a horizontal distance of 24 feet. Find the angle the ramp makes with the ground. Is it less than 5°?

Part 3 · Partner practice

Partner 1

cos A = 0.42. Find m∠A to the nearest tenth of a degree.

Partner 2

In right △ABC, ∠C is the right angle, AC = 8, and BC = 5. Find AB, m∠A, and m∠B. Round to the nearest tenth.

